

1 What is Active Directory (AD)?

Active Directory (AD) is a **Microsoft directory service** used to **manage users, computers, permissions, and resources centrally** in a Windows domain network.

Simple Definition:

👉 *Active Directory is like a digital register (database) that stores information about users, computers, and resources in a network.*

Why Active Directory is used:

- Centralized user management
- Single login (SSO)
- Security and access control
- Policy enforcement
- Scalability for large networks

📌 Real-life example:

College IT admin manages **1000 students** using **one login system** instead of creating users on every PC.

2 What is AD DS (Active Directory Domain Services)?

AD DS is the **core service/role** that provides Active Directory functionality.

AD vs AD DS:

- **Active Directory** → Concept / Directory
- **AD DS** → Actual Windows Server role that runs AD

AD DS provides:

- Authentication (login)
- Authorization (permission)
- User & computer accounts
- Group Policy support

 Without AD DS, **Active Directory cannot function.**

3 What is a Domain?

A **Domain** is a **logical boundary** where:

- Users
- Computers
- Servers

are managed under **one security database.**

 Example:

Domain Name: codingseekho.local

4 What is Organizational Unit (OU)?

An **OU (Organizational Unit)** is a **container inside Active Directory** used to **organize users, computers, and groups.**

Why OU is needed:

- Apply different policies to different departments
- Easy management
- Delegation of control

 Example OU Structure:

codingseekho.local

```
├— Students
├— Faculty
├— HR
└— IT
```

5 What is Group Policy (GPO)?

Group Policy is a feature that allows administrators to **control settings of users and computers centrally.**

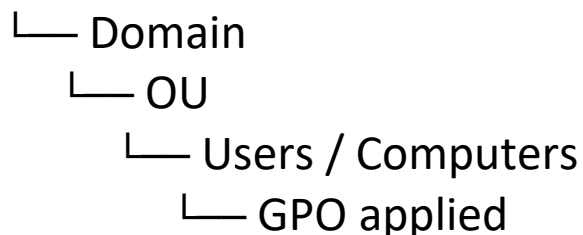
Group Policy can:

- Block USB
- Disable Control Panel
- Enforce password policy
- Set wallpaper
- Restrict software installation

📌 Policies apply via **GPO (Group Policy Object)**.

6 Relationship Between AD, OU and GPO

Active Directory



📌 **Rule:**

👉 GPO is applied to **OU**, not directly to users.

7 Prerequisites to Install AD DS

- Windows Server (2016 / 2019 / 2022)
- Static IP address
- Administrator account

8 How to Install AD DS (Step-by-Step)

Step 1: Open Server Manager

- Click **Manage**
- Select **Add Roles and Features**

Step 2: Select Role

✓ Choose **Active Directory Domain Services**

Step 3: Install

- Click **Next → Install**

Step 4: Promote Server to Domain Controller

- Click **Promote this server to a domain controller**
- Select **Add a new forest**
- Domain name:

codingseekho.local

Step 5: Set Password & Finish

- Set DSRM password
 - Restart server
-  AD DS Installed Successfully

How to Create an OU

Steps:

1. Open **Active Directory Users and Computers**
2. Right-click domain name
3. Click **New → Organizational Unit**
4. Name it:

Students

 Create more OUs:

Faculty

HR

IT

How to Create a User in AD

Steps:

1. Right-click **OU (Students)**
2. New → User

3. Enter:

Username: student1

Password: *****

4. Finish

 User is now part of domain.

1 1 What is GPMC (Group Policy Management Console)?

Used to **create, edit, and link GPOs.**

Open:

Server Manager → Tools → Group Policy Management

1 2 How to Create a GPO

Example 1: Disable Control Panel for Students

Steps:

1. Open **Group Policy Management**
2. Right-click **Students OU**
3. Create a GPO → Name:

Disable_Control_Panel

4. Edit GPO:

User Configuration

→ Administrative Templates

→ Control Panel

→ Prohibit access to Control Panel = Enabled

 Control Panel blocked for students.

Example 2: Disable USB Storage

Computer Configuration

→ Administrative Templates

→ System

→ Removable Storage Access

→ All Removable Storage classes: Deny all access

 Used in labs and offices.

Example 3: Set Desktop Wallpaper

User Configuration

→ Administrative Templates

→ Desktop

→ Desktop Wallpaper

1 3 Link GPO to OU

 Automatically linked when created from OU

OR

Right-click OU → **Link an Existing GPO**

1 4 How to Check GPO Applied

On client PC:

gpupdate /force

Check applied policy:

gpresult /r

1 5 Real-Life Example (College)

Students OU

- USB Disabled
- Control Panel Disabled

Faculty OU

- USB Allowed
- Software Installation Allowed

HR OU

- Access to payroll system

1 6 Interview Questions & Answers

Q: What is Active Directory?

 Centralized directory service for managing users and computers.

Q: Difference between AD and AD DS?

 AD is concept, AD DS is Windows role.

Q: Why OU is used?

 To apply policies and organize users.

Q: Can one GPO apply to multiple OUs?

 Yes.

1 7 Cybersecurity Angle (Important)

- GPO helps prevent malware via:
 - USB blocking
 - Software restriction
 - Password enforcement
- AD logs help in **incident investigation**